

AVANAMÉ

SOPHIA CRM Automation

Client Handover Document

Structure, Writing Guide & SOPHIA-Specific Template

For [Client Name] • Prepared by [Developer Name]

Client	[Client Company Name]
Project	SOPHIA CRM Automation System
Document Version	1.0
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Prepared By	[Developer / Team Name]
Reviewed By	[Avanamé Lead / Project Manager]
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LAYER 1

Executive Overview

Audience: Client / Founder / CEO

1.1 — What SOPHIA Is (Plain Language)

SOPHIA is your fully automated lead management and outreach system. It manages the journey of a potential customer from the moment they enter your ecosystem to the point where they are ready for a direct sales conversation. It works quietly in the background every day — sending emails, following up, reading replies, updating records, and alerting your team only when action is required.

Think of SOPHIA as a disciplined, always-on sales coordinator. When a new lead arrives — whether from your website, a CSV upload, or another intake source — SOPHIA immediately records it, organizes it correctly, and prepares it for outreach. Each weekday morning at 9:00 AM, it sends carefully structured, personalized introduction emails to up to 50 new leads. If someone does not reply, SOPHIA waits 48 hours and sends a follow-up. It will do this a maximum of two times, ensuring persistence without risking your sender reputation.

When a lead replies, SOPHIA reads the response and categorizes it automatically. If someone unsubscribes, it marks them accordingly. If someone expresses interest, SOPHIA flags them as qualified and notifies your team. Only genuinely engaged leads are escalated for human follow-up. Every step is tracked inside your CRM, and your team is notified when real opportunities appear.

Before SOPHIA, this entire process required manual tracking, follow-up reminders, inbox monitoring, and CRM updates. Now, SOPHIA handles this consistently, accurately, and at scale — without fatigue, missed follow-ups, or inconsistent execution.

1.2 — What SOPHIA Does for Your Business

Below are the core business outcomes delivered by SOPHIA. Each outcome reflects a measurable improvement over manual processes.

Business Outcome	What This Means for Your Business
Automated Daily Lead Outreach	Previously, new leads required manual email drafting and sending, often resulting in delays or missed contacts. Now, every weekday at 9:00 AM, up to 50 new leads automatically receive personalized introduction emails. No manual action is required. This ensures consistent, timely first contact without increasing headcount.
Structured Follow-Up Discipline	Previously, follow-ups depended on memory, reminders, or manual tracking — leading to lost opportunities. Now, if a lead does not respond, SOPHIA automatically sends up to two follow-up emails, each spaced 48 hours apart. After two attempts, it stops automatically. This increases response rates while protecting your sender reputation.

Automatic Reply Categorization	Previously, your team had to manually read and interpret every reply. Now, SOPHIA reads incoming responses and categorizes them as interested, unqualified, general inquiry, or unsubscribe. Your team only reviews replies that require human judgment or sales action. This reduces inbox noise and saves operational time daily.
Immediate Qualified Lead Escalation	Previously, interested leads could sit unnoticed in an inbox or spreadsheet. Now, when a lead shows buying intent, SOPHIA automatically marks them as qualified in your CRM and notifies your team. Sales can respond quickly, improving close rates and customer experience.
Centralized CRM Accuracy	Previously, CRM updates required manual data entry and were prone to errors. Now, lead status, follow-up history, reply classification, and contact dates are automatically updated inside your CRM. This ensures accurate reporting and a single source of truth for your pipeline.
Built-In Safeguards & Reputation Protection	Previously, over-contacting or missed unsubscribe requests could damage email reputation. Now, SOPHIA enforces a daily sending cap (50 leads), limits follow-ups to two attempts, and immediately honors unsubscribe requests. This protects your brand and ensures compliant communication practices.

These outcomes collectively transform your outreach process from a manual, inconsistent effort into a structured, automated system that scales predictably while maintaining control and professionalism.

1.3 — The Lead Journey (Visual Lifecycle)

Below is the complete journey a lead follows inside SOPHIA, from first contact to sales handoff and final outcome.

The Lead Lifecycle

Step 1: A new lead enters your business.

This happens either through your website contact form or by uploading a lead file into the system.

Step 2: The system records and organizes the lead.

The lead is stored in your CRM, checked for duplicates (when applicable), and assigned the correct category.

Step 3: An introduction email is sent automatically.

At 9:00 AM on the next business day, the lead receives a personalized outreach email. No manual sending is required.

Step 4: If there is no reply, structured follow-up begins.

After 48 hours, the system sends a follow-up email. If there is still no response, a second and final follow-up is sent 48 hours later. No more than two follow-ups are ever sent.

Step 5: When a reply arrives, it is automatically reviewed.

The system reads the response and determines whether the lead is interested, not interested, requesting information, or unsubscribing.

Step 6: Qualified leads are flagged and prepared for sales review.

If buying intent is detected, the lead is marked as qualified in your CRM and your team receives a notification with full context.

—— AUTOMATION ENDS HERE — HUMAN SALES BEGINS ——

Step 7: A sales associate reviews the qualified lead.

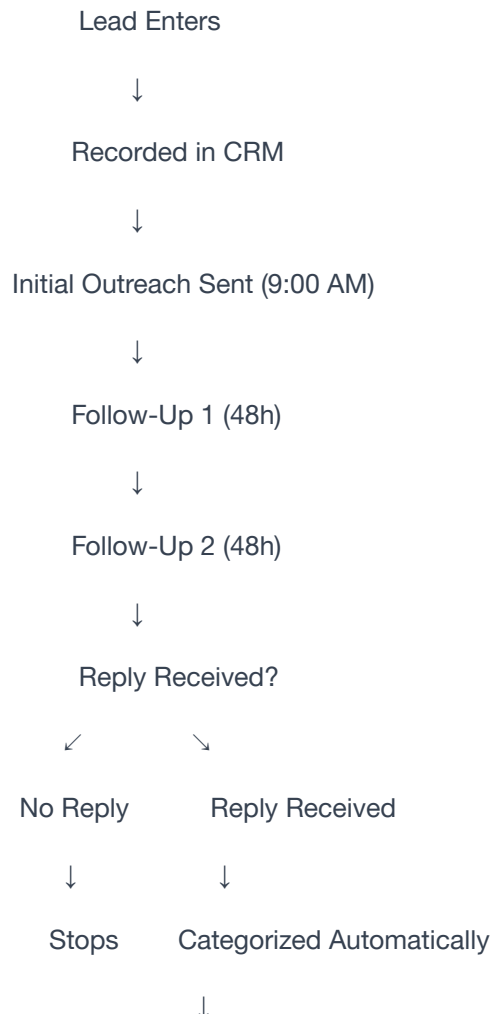
Your team conducts a vetting or discovery conversation to understand needs, budget, and timing.

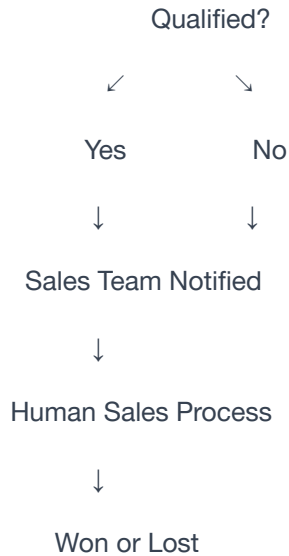
Step 8: The opportunity is progressed.

Based on the conversation, the lead may be marked as Negotiation, Quotation Sent, Closed–Won, or Closed–Lost.

Step 9: The deal is closed.

The final outcome is recorded in your CRM for accurate reporting and future follow-up.

Simple Lifecycle Diagram:



This lifecycle ensures that:

- Every lead receives structured attention.
- No follow-up is missed.
- Your team only engages when real opportunity exists.
- Responsibility clearly transitions from automation to human sales at the correct stage.

1.4 — Your Responsibilities vs. SOPHIA's

Clarity in responsibility ensures nothing is missed. Below is a clear division of what SOPHIA manages automatically and what requires human oversight.

SOPHIA Handles Automatically	Your Team Handles
Sending personalized outreach emails every weekday at 9:00 AM (up to 50 new leads per day).	Uploading new lead files when needed or ensuring website forms remain active.
Sending up to two follow-up emails per lead, each spaced 48 hours apart.	Responding to qualified lead notifications when alerted.
Reading and categorizing every incoming reply (interested, unqualified, general inquiry, unsubscribe).	Conducting vetting or discovery calls with interested prospects.
Marking qualified leads and notifying the sales team immediately.	Preparing and sending quotations after discovery calls.
Updating lead status automatically in your CRM after outreach, replies, and follow-ups.	Updating final deal outcomes (Won or Lost) after a sales decision is made.

Honoring unsubscribe requests immediately and preventing further contact.	Reviewing leads flagged for “manual review” when the system cannot confidently categorize a reply.
Monitoring follow-up limits (maximum two attempts per lead).	Performing a brief weekly CRM review to ensure pipeline accuracy.
Logging system errors and sending immediate alert notifications if something fails.	Reviewing error alert emails when received and escalating to the developer if necessary.
Monitoring quotation aging and closed-lost follow-up reminders automatically.	Taking action on quotation reminders or closed-lost alerts when notified.

Responsibility Transition Point

SOPHIA manages everything from lead intake through qualification.

Human responsibility begins once a lead is marked as qualified and your team is notified.

From that point forward, closing the opportunity — including vetting, pricing, negotiation, and final status updates — is a human responsibility.

1.5 — When Something Goes Wrong

SOPHIA is designed to run automatically and notify you if something fails. However, it is important to recognize warning signs early so that issues can be addressed quickly.

Below are the most common signals you may see, what they typically mean, and what action to take.

What You'll Notice	What It Means	What To Do
No new leads appearing in your CRM after a website submission or file upload	A lead intake process may have paused, failed, or stopped running.	Check whether the lead file was uploaded correctly or whether the website form submission was received. If leads are still missing, contact the developer.
You receive an “Error Alert” email	The system detected a technical issue in one of the automated processes.	Forward the alert email to the developer immediately. Do not attempt manual fixes inside the CRM unless instructed.
Leads report not receiving emails	Outbound email sending may be paused due to a configuration issue, template problem, or account restriction.	Confirm whether recent leads received outreach. If emails appear missing, contact the developer.

Qualified leads are not triggering notifications	The escalation or notification process may not be firing correctly.	Check your CRM for recently marked qualified leads. If no notification was received, contact the developer.
Follow-ups do not appear to be sending	The follow-up scheduling logic may have been interrupted.	Review a few recent “Awaiting Response” leads in the CRM. If follow-ups are missing, contact the developer.

1.6 — Glossary of Terms

Below are the key terms you may encounter in your CRM, notifications, or internal dashboards. Definitions are provided in plain language for clarity.

Term	Definition
Assigned To	The team member responsible for handling the lead once it reaches the sales stage.
Awaiting Response	A lead that has received outreach and is waiting for a reply.
Closed–Lost	A lead that did not convert into a sale after the sales process was completed.
Closed–Won	A lead that successfully converted into a customer.
Date First Contacted	The date the first outreach email was sent to the lead.
Date Lead Responded	The date the lead replied to an outreach email.
Email Template Used	The specific email message version that was sent to the lead.
Follow-Up	An automated reminder email sent when a lead does not respond to the initial outreach.
Follow-Up Count	The number of follow-up emails that have been sent to a lead (maximum of two).
General Inquiry	A reply that asks a question but does not clearly indicate buying intent.
Interest Level	A score (1–5) reflecting how strongly a lead’s reply indicates buying intent.
Lead Owner Notes	Internal notes added by your team to provide context or updates about a lead.

Lead Source	The channel through which the lead entered the system (e.g., website form, uploaded file).
Lead Status	The current stage of a lead in the sales lifecycle (e.g., New, Awaiting Response, Qualified).
Negotiation	A stage indicating that pricing or terms are being discussed with the lead.
New / Uncontacted	A lead that has entered the system but has not yet received outreach.
Qualified – Pending Vetting	A lead identified as interested and awaiting a human discovery or vetting call.
Quotation Sent	A stage indicating that a formal price proposal has been delivered to the lead.
Unqualified	A lead that is not a good fit based on their reply or business criteria.
Unsubscribed	A contact who has requested to stop receiving emails and will not be contacted again.
Vertical	The industry category assigned to a lead, such as Education, Healthcare, or Corporate Events.

LAYER 2

Operational Guide*Audience: Operations Manager / VA / Day-to-Day Operator***2.1 — Daily Operations Checklist**

This checklist should be completed every working day.
Each task should take less than 5 minutes.

The goal is not to “manage” SOPHIA — it runs automatically — but to confirm everything is functioning normally and act quickly if something requires human attention.

✓	Daily Task	Where to Check	What “Normal” Looks Like
☐	Review system error alerts	Gmail inbox (Error Alert emails)	No error alerts received in the last 24 hours
☐	Confirm daily outreach executed	CRM (filter by “Date First Contacted = Today”)	Between 1–50 new leads show today’s contact date
☐	Review qualified lead notifications	Gmail + CRM	All qualified leads visible in CRM and actioned within 24 hours
☐	Review “Needs Manual Review” flags	CRM (filter by Lead Status)	0–2 leads flagged for manual review
☐	Verify follow-ups triggered correctly	CRM (filter by “Awaiting Response”)	Leads older than 48 hours show Follow-Up Count incremented
☐	Confirm intake folder is clear (if uploading CSV files)	Google Drive intake folder	No unprocessed files remaining

2.2 — Standard Operating Procedures

This section contains repeatable procedures for recurring operational tasks.
Each SOP follows the format:

Task Name → When to Perform → Prerequisites → Steps → Verification

These procedures should be followed exactly to maintain system integrity and data accuracy.

SOP 2.2.1 — Uploading a New Lead File (Google Drive Intake)

When to Perform:

Whenever a new batch of leads is ready for outreach (e.g., exported from a data source).

Prerequisites:

- File must be in CSV format (.csv).
 - Each row must contain valid email addresses.
 - File must follow the naming convention:
 - [Vertical]_[Description].csv
 - **Example:** Education_JanuaryBatch.csv
-

Steps

1. Confirm the CSV file contains properly formatted email addresses.
 2. Rename the file according to the required naming convention.
 3. Open Google Drive.
 4. Navigate to the designated lead intake folder.
 5. Upload one file at a time.
 6. Wait 5–10 minutes for processing.
-

Verification

- The file should disappear from the intake folder.
- The file should appear in the archive folder.
- New leads should appear in the CRM with status “New / Uncontacted.”

If the file remains in the intake folder after 15 minutes:

- Check for an error alert email.
 - If no alert is visible, contact the developer.
-

SOP 2.2.2 — Handling a “Needs Manual Review” Lead

When to Perform:

When a lead is flagged for manual review in the CRM.

Prerequisites:

- Access to the CRM.
 - Visibility of the flagged reply.
-

Steps

1. Open the CRM.
 2. Filter by Lead Status = “Needs Manual Review.”
 3. Open the flagged lead record.
 4. Review the reply content in full context.
 5. Determine the correct classification:
 - a. Qualified – Pending Vetting
 - b. Unqualified
 - c. Negotiation
 - d. Unsubscribed
 - 6.
 7. Update the Lead Status accordingly.
 8. Save changes.
-

What You Must NEVER Do

- Do not change the lead status back to “New / Uncontacted.”
- Do not delete the reply history.
- Do not resend outreach emails manually.

Verification

- The lead should no longer appear in the “Needs Manual Review” filter.
 - The updated status should reflect the correct sales stage.
-

SOP 2.2.3 — Responding to a Qualified Lead Notification

When to Perform:

When a “Qualified Lead Alert” email is received.

Response Time Expectation:

Within 4 business hours.

Prerequisites:

- Access to CRM.
 - Access to email notification.
-

Steps

1. Open the qualified lead notification email.
2. Click the CRM link provided.
3. Review:
 - a. Lead Name
 - b. Company
 - c. Original outreach sent
 - d. Reply content

- e. Interest Level
 - f. Event date (if applicable)
 4. Assign the lead to the appropriate sales representative (if not already assigned).
 5. Initiate contact (call or reply email).
 6. After conversation, update Lead Status to:
 - a. Negotiation
 - b. Quotation Sent
 - c. Closed–Won
 - d. Closed–Lost
-

Verification

- Lead Status reflects post-call stage.
 - Notes are added to Lead Owner Notes field.
 - CRM record contains updated timestamps.
-

SOP 2.2.4 — Adding or Editing an Email Template

When to Perform:

When updating messaging or creating a new campaign version.

Prerequisites:

- Access to the Email Templates Google Sheet.
-

Steps

1. Open Google Sheets → Email Templates.
 2. Add a new row or edit an existing one.
 3. Ensure:
 - a. Vertical Name matches CRM vertical exactly.
 - b. Template Name matches subject configuration.
 4. Insert placeholders carefully (e.g., {{First Name}}).
 5. Save changes.
-

Verification

- Template appears in the sheet.
 - No missing placeholder errors occur the next outreach cycle.
-

SOP 2.2.5 — Activating or Deactivating a Subject Line

When to Perform:

When testing new messaging or retiring a subject line.

Prerequisites:

- Access to Email Subjects Google Sheet.
-

Steps

1. Open Google Sheets → Email Subjects.
 2. Locate the desired subject row.
 3. Toggle the “Active” column:
 - a. TRUE = Active
 - b. FALSE = Disabled
 4. Save changes.
-

Verification

- Subject rotation reflects updated Active status.
 - Disabled subjects are no longer used in outreach.
-

SOP 2.2.6 — Handling an Error Alert Email

When to Perform:

Immediately upon receiving an “Error Alert” email.

Prerequisites:

- Access to Gmail.
-

Steps

1. Open the alert email.
 2. Review:
 - a. Workflow name
 - b. Error message
 - c. Timestamp
 3. Do not attempt manual configuration changes.
 4. Forward the email to the developer with a short note.
-

Verification

- Developer acknowledges receipt.
 - Issue resolved confirmation received.
-

SOP 2.2.7 — Weekly System Health Review

When to Perform:

Every Friday before end of business.

Prerequisites:

- Access to CRM.
 - Access to Google Sheets (Error Logs).
-

Steps

1. Review CRM for:
 - a. Leads stuck in “Awaiting Response” beyond 7 days.
 - b. Leads with Follow-Up Count = 2 but no further status.
 2. Review Error Logs sheet for recurring patterns.
 3. Review subject “Used Count” distribution.
 4. Confirm no backlog of manual review flags.
-

Verification

- No abnormal lead stagnation.
 - No recurring error pattern.
 - Subject usage balanced.
-

SOP 2.2.8 — Uploading Leads via Zap Intake to Notion Folder

When to Perform:

When using the alternate Zap-based intake process.

Prerequisites:

- CSV formatted correctly.
 - Email and phone fields validated.
-

Steps

1. Rename file according to vertical naming convention.
 2. Open Google Drive.
 3. Navigate to “Zap Intake to Notion” folder.
 4. Upload one file.
 5. Wait up to 10 minutes.
-

Verification

- File moves automatically to the archive folder.
- Leads appear in CRM with status “New / Uncontacted.”
- No duplicate entries created.

If duplicates appear or file remains in intake folder:
Contact the developer immediately.

2.3 — Understanding Gmail Labels

SOPHIA uses Gmail labels to clearly signal when attention is required and when no action is needed.

These labels help you quickly distinguish between routine system activity and items that require human follow-up.

The table below explains each label, what triggers it, and what action (if any) is required.

Gmail Label	What It Means	Action Required
Sales Alert	A lead has replied with buying intent and has been marked as Qualified – Pending Vetting in the CRM.	Open the linked CRM record immediately. Begin vetting or discovery within 4 business hours. Update lead status after contact.
Manual Review	A lead reply could not be confidently categorized by the system and requires human review.	Open the CRM record, read the reply in full context, and update the Lead Status appropriately. Do not resend outreach emails.
Alert	A template issue, missing placeholder, subject configuration problem, or related setup issue was detected during processing.	Review the alert details carefully. If the issue is not obvious, forward the alert to the developer. Do not modify system configurations unless instructed.
Error	A workflow execution failure occurred. This means a technical process did not complete successfully.	Forward the email to the developer immediately. Do not attempt to manually correct CRM data unless specifically advised.
Follow-Up Sent (if present)	A scheduled follow-up email was successfully sent to a lead who had not replied.	No action required. This is informational only.
Website Form Submission (if used)	A new lead was submitted through the website contact form and has been automatically added to the CRM.	No action required unless urgent. Review CRM if immediate follow-up is needed.

Qualified — Negotiation (if applied)	A previously qualified lead has progressed to negotiation or quotation stage.	Ensure sales process is active. Monitor deal progress and update CRM accordingly.
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Important Notes

- Sales Alert and Manual Review labels always require human attention.
- Alert and Error labels should be escalated promptly to prevent workflow disruption.
- Informational labels (such as Follow-Up Sent or Website Submission) do not require action unless unusual patterns are observed.

If you are unsure whether a label requires action, open the email and review the context. When in doubt, escalate to the developer rather than making system changes.

2.4 — Understanding Lead Statuses

Lead Status is the most important field in your CRM.

It determines where a lead is in the lifecycle, what automation (if any) will run next, and whether human action is required.

The table below maps every active status used in SOPHIA to:

- What it means in plain language
- What triggers the status change
- What happens next

Lead Status	Meaning	Triggered By	What Happens Next
New / Uncontacted	A lead has entered the system but has not yet received outreach.	Website form submission, CSV upload, or manual intake.	SOPHIA schedules the lead for the next weekday 9:00 AM outreach batch.
Awaiting Response	The first outreach email has been sent and the system is waiting for a reply.	Daily outreach automation.	If no reply, SOPHIA sends up to two follow-ups spaced 48 hours apart. If a reply arrives, it is automatically classified.
Qualified – Pending Vetting	The lead has shown buying interest and requires human follow-up.	Automatic reply classification detecting positive intent.	Sales team is notified. A discovery or vetting call should occur within 4 business hours.
Negotiation	The lead is actively discussing pricing, terms, or next steps with sales.	Sales team updates status after discovery call.	Opportunity progresses toward quotation or closure. No further automated outreach occurs.

Quotation Sent	A formal quote has been delivered to the lead.	Sales team updates status after sending quotation.	The system monitors for aging quotations and may trigger internal reminders if the event date is approaching.
Closed–Won	The lead converted into a paying customer.	Sales team updates status after deal closure.	Automation stops permanently for this lead. Record remains for reporting.
Closed–Lost	The opportunity did not convert after sales review.	Sales team updates status after decision.	Automation stops. Internal monitoring may remind the team after the event date for follow-up analysis.
Unqualified	The lead is not a good fit based on response, budget, or criteria.	Automatic reply classification or manual sales decision.	All outreach stops permanently.
Unsubscribed	The lead requested no further contact.	Automatic reply classification detecting unsubscribe intent.	All future emails are permanently blocked.
Needs Manual Review (if used as a status)	The system could not confidently interpret the reply.	Reply classification uncertainty.	Operator must review the reply and manually update the Lead Status.
Transferred to Notion (legacy sheet flow only)	A qualified lead was moved from the sheet-based system into the CRM.	Sales escalation workflow.	Lead continues through standard CRM lifecycle stages.

How Status Drives Automation

- Only leads marked **New / Uncontacted** are eligible for initial outreach.
- Only leads marked **Awaiting Response** are eligible for follow-ups.
- Once a lead becomes **Qualified – Pending Vetting**, automation pauses and human sales begins.
- Any status marked **Unqualified, Unsubscribed, Closed–Won, or Closed–Lost** permanently stops automated emails.

Important Operational Rule

Never change a lead's status back to **New / Uncontacted** after outreach has already been sent. Doing so may re-trigger automated emails and create duplicate communication.

If you are unsure which status to select, refer to this table first. If still unclear, escalate to the developer or sales manager before making changes.

2.5 — Escalation Matrix

This matrix defines when escalation is required, who should be contacted, through which channel, and within what timeframe.

Escalation should occur whenever an issue affects lead processing, outreach delivery, data integrity, or sales notifications. The objective is to prevent revenue loss and avoid manual workarounds that may introduce further inconsistencies.

Situation	Escalate To	Method	Timeframe
Error Alert email received from SOPHIA	Automation Developer	Forward email + brief context via Email	Within 1 hour
No outreach emails sent for 2 consecutive business days	Automation Developer	Direct message (WhatsApp/Slack) + Email	Immediately
Leads remain in “New / Uncontacted” for more than 48 hours	Automation Developer	Email with CRM screenshot	Within 2 hours
Follow-ups not sending after 48-hour threshold	Automation Developer	Email with affected Lead IDs	Within 2 hours
Qualified lead marked in CRM but no Sales Alert received	Automation Developer	Direct message + Email	Within 2 hours
Website form submissions not appearing in CRM	Automation Developer	Direct message	Immediately
CSV file remains in intake folder for more than 15 minutes	Automation Developer	Direct message	Immediately
Duplicate leads detected in CRM	Automation Developer	Email with duplicate record links	Same business day

Lead complaint about unwanted emails	Project Manager + Automation Developer	Email with full details and screenshots	Within 30 minutes
Gmail account warning (rate limit or suspicious activity)	Automation Developer	Direct message	Immediately
Unclear how to classify a lead reply	Project Manager	Email	Before taking action
CRM field appears missing or renamed	Automation Developer	Email	Same business day
System-wide inactivity (no outreach, no follow-ups, no classifications)	Automation Developer	Direct call + message	Immediately

Escalation Principles

- Do not attempt to modify workflow logic without developer approval.
- Do not manually change automation-driving fields (Lead Status, Follow-Up Count, Assigned To) unless instructed.
- Always include screenshots, timestamps, and lead names when escalating.
- If revenue-impacting, notify the Sales Manager in parallel.

Defined Roles (To Be Completed Before Final Delivery)

- **Automation Developer:** [Name] – [Email] – [Direct Contact Channel]
- **Project Manager:** [Name] – [Email]
- **Sales Manager:** [Name] – [Direct Contact Channel]

LAYER 3

Technical Reference

Audience: Developer / Technical Maintainer

3.1 — System Architecture Overview

SOPHIA is a multi-platform automation architecture built across:

- n8n (primary orchestration engine)
- Zapier (event-based ingestion)
- Notion (primary CRM)
- Google Sheets (configuration + logging layer)
- Google Drive (file intake + archival)
- Gmail (outbound + inbound communication + labeling)
- Webflow (website form capture)
- Clay (lead source export tool)

3.1.1 — System Components

Component	Role	Instance URL / Location	Access Credentials
n8n	Primary automation engine for outreach, follow-ups, reply classification, and error logging.	Production n8n instance (self-hosted or cloud URL)	Stored in credential vault (n8n credentials manager)
Zapier	Event-driven ingestion from Drive and Webflow into Notion.	Zapier dashboard (production workspace)	Zapier account vault
Google Drive	Stores intake CSV files and archival folders.	Drive → Lead Intake folders	Google OAuth2 credential
Google Sheets	Stores email subjects, email templates, follow-up usage counters, and error logs.	Production Sheets file(s)	Google OAuth2 credential
Gmail	Sends outreach, follow-ups, notifications, and reads replies.	Primary Gmail account	OAuth2 token stored in n8n
Notion CRM	Master database for all lead records and lifecycle tracking.	Notion workspace → Leads & Customers DB	Notion integration token
Webflow	Captures inbound website leads via webhook.	Webflow form → Zapier webhook	Webflow site access

Clay	Source system for exporting cold lead CSV files.	Clay dashboard	Clay account
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3.1.2 — Data Authority Map

This resolves the Google Sheets vs. Notion ambiguity.

Data Domain	Authoritative System	Sync Direction	Why This Architecture
Lead Master Records	Notion	Inbound only	Notion is the CRM system of record; Sheets is not used as CRM in production.
Email Activity Tracking	Notion	n8n → Notion	Outreach and follow-up timestamps stored in CRM for reporting integrity.
Reply Classification	Notion	Gmail → n8n → Notion	Classification updates written directly to CRM; no duplication.
Follow-Up Tracking	Notion	n8n → Notion	Follow-Up Count and Next Follow-Up Date maintained in CRM.
Qualified Lead Records	Notion	Direct write	Single source of truth avoids dual-CRM inconsistency.
Email Templates / Subjects	Google Sheets	Read-only into n8n	Sheets used as flexible configuration store for marketing team.
Subject Usage Counters	Google Sheets	n8n → Sheets	Keeps rotation fairness without polluting CRM.
Error Logs	Google Sheets	n8n → Sheets	Lightweight logging mechanism without requiring separate logging stack.

3.2 — Workflow Registry & Configuration

This section provides a standardized technical reference card for each SOPHIA workflow.

Each card documents:

- Trigger configuration
- Data sources
- Filtering logic
- Field updates
- Error handling
- Cross-workflow dependencies
- Design rationale

Workflow 1 — CSV Import to Google Sheets

Field	Value
Workflow Name	SOPHIA – CSV Import to Google Sheets
n8n Workflow ID	[INSERT n8n WORKFLOW ID HERE]
Trigger Type	Manual Trigger
Schedule	N/A
Timezone	N/A
Daily Send Limit	N/A
Data Source	Google Drive → CSV_Imports Folder
Filter Criteria	contentType = text/csv
Template Selection Logic	N/A
Post-Send Updates	Appends rows to Master Leads Sheet; Moves file to Archive folder
Error Handling	Error Logger workflow triggered; no retry logic
Dependencies	Google Drive, Google Sheets
Design Decision Notes	Manual trigger ensures controlled ingestion; numeric filename sorting ensures deterministic batch order.

Workflow 2 — Email Outreach (Google Sheets-Based)

Field	Value
Workflow Name	SOPHIA – Email Outreach (Sheets CRM)
n8n Workflow ID	[INSERT ID]
Trigger Type	Cron
Schedule	0 9 * * 1-5
Timezone	America/Toronto (or production timezone)
Daily Send Limit	50
Data Source	Google Sheets → Lead Status = "New"

Filter Criteria	Lead Status = New AND Valid Email Regex
Template Selection Logic	Round-robin by Used Count; Vertical match; Active = TRUE
Post-Send Updates	Lead Status → Contacted; Date First Contacted; Subject Used Count increment
Error Handling	Missing template → alert email; Placeholder error → flagged
Dependencies	Email Subjects Sheet; Email Templates Sheet
Design Decision Notes	50 cap protects sender reputation; 9 AM aligns with business open hours.

Workflow 3 — Reply Classification (Sheets-Based)

Field	Value
Workflow Name	SOPHIA – Reply Classification (Sheets CRM)
n8n Workflow ID	[INSERT ID]
Trigger Type	Cron
Schedule	0 0,6,15 * * *
Timezone	Production timezone
Daily Send Limit	N/A
Data Source	Gmail (Unread Messages)
Filter Criteria	Lead Status = Contacted
Template Selection Logic	Rule-based keyword classification
Post-Send Updates	Lead Status updated; Interest Level assigned
Error Handling	Unmatched replies flagged for manual review
Dependencies	Gmail; Google Sheets
Design Decision Notes	Multiple daily sweeps reduce response latency.

Workflow 4 — Follow-Up Automation (Sheets-Based)

Field	Value
-------	-------

Workflow Name	SOPHIA – Follow-Up Automation (Sheets CRM)
n8n Workflow ID	[INSERT ID]
Trigger Type	Cron
Schedule	0 14 * * 1-5
Timezone	Production timezone
Daily Send Limit	50
Data Source	Google Sheets → Lead Status = Contacted
Filter Criteria	Date First Contacted > 48h; Follow-Up Count < 2
Template Selection Logic	Vertical-based template selection
Post-Send Updates	Follow-Up Count increment; Date Follow-Up Sent
Error Handling	Missing template → alert
Dependencies	Email Templates Sheet
Design Decision Notes	48-hour spacing chosen to balance persistence with reputation safety.

Workflow 5 — Notion Transfer (Sales Escalation)

Field	Value
Workflow Name	SOPHIA – Qualified Lead Transfer
n8n Workflow ID	[INSERT ID]
Trigger Type	Cron
Schedule	* / 10 * * * *
Timezone	Production timezone
Daily Send Limit	N/A
Data Source	Google Sheets → Status = Qualified--Pending Vetting
Filter Criteria	Lead Status = Qualified--Pending Vetting
Template Selection Logic	N/A
Post-Send Updates	Create Notion page; Update Sheet status to Transferred

Error Handling	Alert on Notion API failure
Dependencies	Notion API; Gmail
Design Decision Notes	10-minute interval ensures near real-time sales handoff.

Workflow 6 — Drive Intake to Notion (Zapier)

Field	Value
Workflow Name	SOPHIA – Drive Intake (Zapier)
Platform	Zapier
Trigger Type	New File in Folder
Daily Send Limit	500 rows per file
Data Source	Google Drive → Zap Intake Folder
Filter Criteria	File extension = CSV
Template Selection Logic	Deduplicate by Email OR Phone
Post-Send Updates	Create Notion record; Archive file
Error Handling	Zap error alert
Dependencies	Notion; Google Drive
Design Decision Notes	OR-based dedupe prevents duplicate records.

Workflow 7 — Webflow Contact Form to Notion (Zapier)

Field	Value
Workflow Name	SOPHIA – Webflow Inbound
Platform	Zapier
Trigger Type	Webhook
Filter Criteria	Topic-based branching
Post-Send Updates	Create Notion record; Send confirmation email
Error Handling	Zapier alert
Dependencies	Webflow; Notion; Gmail

Design Decision Notes	Topic-based qualification accelerates sales response.
------------------------------	---

Workflow 8 — Notion CRM Email Outreach (Primary Engine)

Field	Value
Workflow Name	SOPHIA – Notion Email Outreach
n8n Workflow ID	[INSERT ID]
Trigger Type	Cron
Schedule	0 9 * * 1-5
Timezone	Production timezone
Daily Send Limit	50
Data Source	Notion → Lead Status = New / Uncontacted
Filter Criteria	Assigned To = SOPHIA; Vertical not empty
Template Selection Logic	Round-robin; Vertical match; Active = TRUE
Post-Send Updates	Lead Status → Awaiting Response; Date First Contacted
Error Handling	Missing subject/template → alert
Dependencies	Gmail; Google Sheets; Notion
Design Decision Notes	Notion as source-of-truth CRM eliminates dual-write risk.

Workflow 9 — Notion Reply Classification

Field	Value
Workflow Name	SOPHIA – Notion Reply Classification
n8n Workflow ID	[INSERT ID]
Trigger Type	Cron
Schedule	0 15,18 * * *
Timezone	Production timezone
Data Source	Gmail (Unread)
Filter Criteria	Lead Status = Awaiting Response

Template Selection Logic	Rule-based keyword classifier
Post-Send Updates	Lead Status update; Interest Level; Date Lead Responded
Error Handling	Manual Review flag
Dependencies	Gmail; Notion
Design Decision Notes	Deterministic classification avoids AI unpredictability.

Workflow 10 — Notion Follow-Up Automation

Field	Value
Workflow Name	SOPHIA – Notion Follow-Up Engine
n8n Workflow ID	[INSERT ID]
Trigger Type	Cron
Schedule	0 14 * * 1-5
Timezone	UTC
Daily Send Limit	50
Data Source	Notion → Lead Status = Awaiting Response
Filter Criteria	Follow-Up Count < 2; 48h elapsed
Template Selection Logic	Vertical-based template
Post-Send Updates	Follow-Up Count increment; Next Follow-Up Date set
Error Handling	Template validation alerts
Dependencies	Gmail; Google Sheets
Design Decision Notes	Hard 2-follow-up cap protects sender health.

Workflow 11 — Error Logger

Field	Value
Workflow Name	SOPHIA – Global Error Logger
n8n Workflow ID	[INSERT ID]
Trigger Type	Error Trigger/Schedule/Event-driven

Data Source	n8n execution failures
Post-Send Updates	Append to Error Logs Sheet; Send Gmail alert
Dependencies	Google Sheets; Gmail
Design Decision Notes	Centralized logging prevents silent automation failures.

3.3 — Reply Classification Logic (Full Rules)

3.3.1 Overview

The SOPHIA Reply Classification engine is a deterministic, rule-based system responsible for interpreting inbound email responses and updating the Lead Status within the CRM.

The system evaluates each inbound reply using a strict priority-based rule sequence. Classification is performed using case-insensitive keyword matching against the normalized email content (subject and body). Once a rule condition is satisfied, evaluation terminates and the corresponding classification is applied.

This design ensures:

- Predictable behavior
- Auditability
- Compliance protection
- Safe automation boundaries

No machine learning or probabilistic inference is used.

3.3.2 Evaluation Order

Inbound replies are evaluated in the following strict sequence:

- Unsubscribe / Do Not Contact
- Unqualified (Explicit Rejection)
- Very Short / Low-Context Reply
- Positive / Buying Intent
- General Inquiry
- Ambiguous / May Be Unqualified
- Fallback (Needs Manual Review)

This order is mandatory. Higher categories override lower categories. For example, if a reply contains both an unsubscribe instruction and an expression of interest, the unsubscribe rule will prevail.

3.3.3 Classification Categories

A. Unsubscribe / Do Not Contact

Purpose: Ensure regulatory compliance and immediate cessation of contact.

Trigger Condition:

The reply contains any of the following phrases:

- unsubscribe
- remove me
- remove us
- stop emailing
- stop contacting
- take me off
- opt out
- do not contact
- please don't email

Resulting CRM Update:

- Lead Status → Unsubscribed
- Interest Level → 0

System Behavior:

- All future outreach is permanently blocked.
- Follow-up scheduling is terminated.
- No sales notification is triggered.

Design Justification:

Compliance and sender reputation protection override all other business logic.

B. Unqualified (Explicit Rejection)

Purpose: Identify leads that have clearly declined interest.

Trigger Condition:

The reply contains phrases such as:

- not interested
- no thanks
- no thank you
- not a fit
- not relevant
- not looking
- no budget
- already have a supplier
- we use someone else

Resulting CRM Update:

- Lead Status → Unqualified
- Interest Level → 0

System Behavior:

- Follow-ups are cancelled.
- No sales alert is sent.

Design Justification:

Explicit rejection indicates no further engagement is appropriate.

C. Very Short / Low-Context Replies

Purpose: Prevent unsafe interpretation of ambiguous short replies.

Trigger Condition:

- Reply length is fewer than 10 characters
OR
- Reply consists solely of minimal acknowledgments such as:
 - Ok
 - thanks
 - thank you
 - noted

Resulting CRM Update:

- Lead Status → Needs Manual Review
- Interest Level → 1

System Behavior:

- Follow-ups are paused.
- No sales notification is triggered.
- Manual review required.

Design Justification:

Low-context replies cannot reliably indicate intent.

D. Positive / Buying Intent

Purpose: Identify explicit commercial interest.

Trigger Condition:

The reply includes phrases indicating intent to purchase or discuss further:

- interested

- let's talk
- book a call
- schedule a call
- can we discuss
- send pricing
- send quote
- what are your rates
- what is the cost
- available next week
- we need this

Resulting CRM Update:

- Lead Status → Qualified – Pending Vetting
- Interest Level → 5

System Behavior:

- Follow-ups stop immediately.
- Sales Alert email is triggered.

Design Justification:

High buying intent requires rapid human response.

E. General Inquiry

Purpose: Capture leads expressing curiosity without explicit purchase signals.

Trigger Condition:

The reply includes inquiry-based language such as:

- can you explain
- how does this work
- tell me more
- what do you offer
- can you share details
- do you provide

Resulting CRM Update:

- Lead Status → Qualified – Pending Vetting
- Interest Level → 3

System Behavior:

- Sales Alert triggered.
- Automation pauses pending human review.

Design Justification:

Inquiry responses may convert with clarification.

F. Ambiguous / May Be Unqualified

Purpose: Identify responses that defer decision-making.

Trigger Condition:

The reply includes timing or uncertainty phrases:

- maybe later
- not right now
- circle back
- next quarter
- currently evaluating
- possibly
- depends

Resulting CRM Update:

- Lead Status → Negotiation
- Interest Level → 2

System Behavior:

- Sales Alert triggered.
- Automation halts until human action.

Design Justification:

Ambiguous responses require manual interpretation.

G. Fallback — Needs Manual Review

Purpose: Safe termination state when no rule matches.

Trigger Condition:

No keywords matched in prior categories.

Resulting CRM Update:

- Lead Status → Needs Manual Review
- Interest Level → 1

System Behavior:

- No automatic notification.
- Manual classification required.

Design Justification:

Prevents incorrect automated classification.

3.3.4 Edge Case Handling

Auto-Reply Detection

Replies containing:

- out of office
- automatic reply
- away from the office
- on vacation

System Behavior:

- Lead Status remains Awaiting Response.
 - Follow-up schedule continues.
-

Bounce / Delivery Failure

Replies originating from mailer-daemon or containing:

- delivery status notification
- undeliverable

System Behavior:

- Lead Status → Unqualified
 - Outreach stops.
-

Multi-Intent Responses

If multiple categories match, the system applies the highest-priority rule.

Example:

“Not interested right now, maybe next quarter.”
→ Classified as Unqualified due to explicit rejection.

Unsubscribe Combined with Interest

Unsubscribe classification always overrides buying intent.

Empty Replies with Attachments

Automatically classified as Needs Manual Review.

3.3.5 Architectural Rationale

The rule-based classification engine was chosen for:

- Full transparency
- Auditability
- Deterministic behavior
- Compliance safety
- Ease of maintenance

This approach ensures that any developer inheriting the system can clearly trace classification decisions and safely extend rule sets without introducing ambiguity.

3.4 — Database Schema Reference

This section documents the complete schema for:

1. The Master Leads Google Sheet (legacy Sheets-based CRM workflows)
2. The Notion CRM – Leads & Customers Database (primary production CRM)

For each field, this section defines:

- Column / Field Name
- Data Type
- Who writes or updates the field (SOPHIA or Human)
- Which workflows reference it
- Whether it drives workflow logic
- Whether it is used as an email placeholder

Fields that drive automation behavior are explicitly flagged.

3.4.1 Master Leads Sheet Schema (Google Sheets)

Used in legacy and configuration-based workflows (Workflow 1–4, 9, 11).

Not authoritative for production CRM state once Notion is active.

Column Name	Data Type	Written By	Used In Workflows	Notes
First Name	Text	CSV Import / Zapier	WF 2, 4	Email placeholder: {{First Name}}
Last Name	Text	CSV Import / Zapier	WF 2, 4	Email placeholder
Email	Text	CSV Import / Zapier	WF 2, 3, 4	Required for outreach. Regex validated before send.

Phone	Text	CSV Import / Zapier	WF 6	Used for deduplication (Zap intake).
Company Name	Text	CSV Import / Zapier	WF 2	Email placeholder
Vertical	Enum (Text)	CSV Import	WF 2, 4	Drives template selection logic.
Location	Text	CSV Import	WF 2	Optional personalization field.
Lead Source	Text	CSV Import / Webflow	WF 6, 7	Reporting field.
Lead Status	Enum	SOPHIA (All WFs)	WF 2, 3, 4, 9	PRIMARY WORKFLOW DRIVER. Never manually reset to “New.”
Date First Contacted	Date	SOPHIA (WF 2)	WF 4	Determines follow-up eligibility (48-hour logic).
Follow-Up Count	Integer	SOPHIA (WF 4)	WF 4	Logic driver. Max value = 2.
Next Follow-Up Date	Date	SOPHIA (WF 4)	WF 4	Determines next send eligibility.
Date Lead Responded	Date	SOPHIA (WF 3, 9)	WF 9	Used for reporting.
Interest Level	Integer (1–5)	SOPHIA	WF 3, 9	Derived from reply classification.
Email Template Used	Text	SOPHIA	WF 2, 4	Reporting and debugging.
Subject Used	Text	SOPHIA	WF 2	Tracks subject rotation.
Assigned To	Text	Human	WF 8	Determines eligibility for automation.
Notes	Text	Human	None	Free-text internal comments.
Event Date	Date	CSV Import / Webflow	WF 10	Used for quotation monitoring alerts.

Quantity	Number	CSV Import / Webflow	WF 7	Sales reference field.
Topic	Text	Webflow	WF 7	Branch logic (General Inquiry vs Event).
Transferred to Notion	Boolean	SOPHIA (WF 5)	WF 5	Prevents duplicate transfer.
Error Flag	Boolean	SOPHIA (WF 11)	WF 11	Indicates workflow failure occurred.

Master Sheet Logic Drivers

The following fields directly control automation:

- Lead Status
- Date First Contacted
- Follow-Up Count
- Next Follow-Up Date
- Vertical
- Assigned To

These fields must not be edited during active automation cycles unless corrective action is required.

3.4.2 Notion CRM Database Schema (Production Authority)

This is the authoritative system for all active lead lifecycle tracking.

Field Name	Data Type (Notion)	Written By	Used In Workflows	Notes
First Name	Rich Text	Intake / SOPHIA	WF 8	Email placeholder
Last Name	Rich Text	Intake	WF 8	Email placeholder
Email	Email	Intake	WF 8, 9, 10	Required for outreach
Phone	Phone	Intake	WF 6	Deduplication field
Company Name	Rich Text	Intake	WF 8	Placeholder
Vertical	Select	Intake	WF 8, 10	Template selection driver
Lead Status	Select	SOPHIA / Human	ALL	PRIMARY LIFECYCLE DRIVER

Assigned To	Select	Human	WF 8	Only leads assigned to “SOPHIA” are eligible for automation
Date First Contacted	Date	SOPHIA	WF 10	Drives 48-hour follow-up logic
Follow-Up Count	Number	SOPHIA	WF 10	Max = 2
Next Follow-Up Date	Date	SOPHIA	WF 10	Determines next eligibility
Date Lead Responded	Date	SOPHIA	WF 9	Reporting
Interest Level	Number (1–5)	SOPHIA	WF 9	Derived from classification
Email Template Used	Text	SOPHIA	WF 8	Debugging & analytics
Subject Used	Text	SOPHIA	WF 8	Rotation tracking
Event Date	Date	Intake	WF 10	Used in aging alerts
Quantity	Number	Intake	WF 7	Sales context
Topic	Select	Intake	WF 7	Branch logic
Quotation Sent Date	Date	Human	WF 10	Used for quotation reminders
Closed Outcome	Select	Human	None	Values: Closed–Won, Closed–Lost
Created Time	System	Notion	Reporting	Auto-generated
Last Edited Time	System	Notion	Reporting	Auto-generated

Fields Used as Email Placeholders

The following CRM fields are dynamically rendered inside email templates:

- First Name
- Last Name
- Company Name
- Vertical
- Event Date

- Quantity

If a placeholder field is empty during rendering, the workflow triggers a template validation alert.

Fields Driving Workflow Logic (Critical)

The following fields directly control automation:

- Lead Status
- Assigned To
- Date First Contacted
- Follow-Up Count
- Next Follow-Up Date
- Vertical

Any manual modification to these fields can alter automation behavior.

Data Authority Clarification

- Notion CRM is the authoritative source of truth for all active leads.
 - Google Sheets stores configuration (templates, subjects) and logs.
 - Lead lifecycle state must never be managed in multiple systems simultaneously.
-

Schema Governance Rules

1. Do not rename workflow-driving fields without updating all workflow nodes.
 2. Do not change Select field values (e.g., Lead Status labels) without updating filter logic.
 3. Do not delete fields used in placeholder rendering.
 4. Any schema change must be regression-tested across:
 - a. Outreach workflow
 - b. Follow-up workflow
 - c. Reply classification
 - d. Error logging
-

3.5 — Integration & Webhook Details

This section documents all external integrations used by SOPHIA, including:

- Webhook endpoints
- API connections
- OAuth configurations
- Credential storage
- Credential rotation procedures

No raw credentials are included in this document. All authentication details are stored in secure credential managers.

3.5.1 Integration Overview

SOPHIA integrates with the following external systems:

- Gmail
- Notion
- Google Sheets
- Google Drive
- Webflow
- Zapier
- Clay

Each integration is documented below with trigger behavior, data exchange, authentication method, and credential governance.

3.5.2 Gmail Integration

Purpose

- Send outreach emails
- Send follow-up emails
- Send qualified lead notifications
- Send error alerts
- Read inbound replies
- Apply Gmail labels

Trigger Direction

- Outbound: SOPHIA → Gmail
- Inbound: Gmail → SOPHIA

Data Sent

Outbound emails include:

- To (Lead Email)
- Subject
- HTML Body
- Labels (Sales Alert, Error, Manual Review, etc.)

Inbound data retrieved:

- Sender
- Subject
- Email body

- Thread ID
- Timestamp

Authentication Method

OAuth2 connection via Google account.

Credential Storage

Stored in:

- n8n Credential Manager → “Gmail Production OAuth”
- Encrypted within automation platform

If Credentials Are Compromised

1. Revoke app access from Google Account → Security → Third-party apps.
 2. Delete Gmail credential inside n8n.
 3. Reconnect using OAuth2 login.
 4. Test send and reply-read workflow.
-

3.5.3 Notion API Integration

Purpose

- Create new lead records
- Update Lead Status
- Update Follow-Up Count
- Update Date First Contacted
- Update Interest Level
- Retrieve eligible leads for outreach

Trigger Direction

- SOPHIA → Notion
- Zapier → Notion

Data Sent

Typical write operations include:

- Lead Status
- Assigned To
- Date First Contacted
- Follow-Up Count
- Interest Level

Authentication Method

Notion Internal Integration Token.

Credential Storage

Stored in:

- Secure Vault → “Notion Production Token”
- n8n Credential Manager
- Zapier app connection

Credential Rotation Procedure

1. Go to Notion → Settings → Integrations.
 2. Revoke existing integration token.
 3. Generate new token.
 4. Update token in n8n and Zapier.
 5. Execute test workflow.
-

3.5.4 Google Sheets Integration

Purpose

- Read Email Templates
- Read Email Subjects
- Increment Subject Used Count
- Append Error Logs

Trigger Direction

SOPHIA → Google Sheets

Data Sent

- Template reads (no write)
- Used Count increment
- Error log rows appended

Authentication Method

OAuth2 via Google account.

Credential Storage

Stored in:

n8n Credential Manager → “Google Sheets Production”

Credential Rotation

1. Revoke app access in Google account.

2. Reconnect OAuth2 in n8n.
3. Test template read and log append.

3.5.5 Google Drive Integration

Purpose

- Monitor CSV intake folders
- Move processed files to archive
- Trigger Zapier ingestion

Trigger Direction

- Google Drive → Zapier
- Google Drive → SOPHIA

Data Sent

- File metadata
- CSV file contents

Authentication Method

OAuth2 via Google account.

Credential Storage

Stored in:

- n8n Credential Manager
- Zapier app connection

If Compromised

1. Revoke Google Drive app access.
2. Reconnect OAuth2 credentials.
3. Upload test CSV file.

3.5.6 Webflow Webhook Integration

Purpose

Capture website contact form submissions and forward them to Zapier.

Trigger Direction

Webflow → Zapier (Webhook Catch Hook)

Webhook URL

Configured inside Zapier trigger step.

Unique URL generated by Zapier and stored in:

- Zapier Dashboard → Trigger Step → Webhook URL

This URL is configured inside Webflow form settings.

Data Received

JSON payload containing:

- First Name
- Last Name
- Email
- Phone
- Company
- Event Date
- Quantity
- Topic

Authentication Method

Security via unique webhook URL (no additional signature validation configured).

If Compromised

1. Regenerate webhook in Zapier.
2. Replace webhook URL inside Webflow form settings.
3. Submit test form.

3.5.7 Zapier Integration

Purpose

- Drive CSV → Notion
- Webflow → Notion

Authentication

Zapier-managed OAuth connections to:

- Google Drive
- Notion
- Gmail

Credential Storage

Zapier account vault (encrypted).

Credential Rotation

1. Reset Zapier account password.
2. Reconnect all app connections.
3. Test triggers.

3.5.8 Clay (Lead Source)

Purpose

Export cold lead CSV files for manual upload.

Integration Type

No API integration

Manual CSV export → Uploaded to Google Drive intake folder.

Authentication

Clay platform login.

Credential Governance

Managed within Clay account security settings.

3.5.9 Credential Governance Policy

- No API keys are hardcoded in workflows.
- All OAuth credentials are encrypted in automation platform.
- No credentials are stored in Google Sheets or Notion.
- Access restricted to authorized administrators only.

3.5.10 Common Authentication Failure Indicators

Symptom	Likely Cause
Outreach emails not sending	Gmail OAuth expired
Leads not updating in CRM	Notion token invalid

Templates not loading	Google Sheets OAuth issue
CSV files not processing	Drive OAuth failure
Website leads not appearing	Webhook URL invalid

3.6 — Recovery & Troubleshooting Playbook

This section provides structured diagnostic and recovery procedures for the most common SOPHIA failure scenarios.

Each scenario follows the format:

- Symptom
- Possible Causes
- Diagnosis Steps
- Fix
- Verification
- Prevention

All procedures assume access to:

- n8n production instance
- Zapier dashboard
- Google Sheets (Templates, Subjects, Error Logs)
- Google Drive intake folders
- Notion CRM

3.6.1 — CSV Import Failure (File Stuck in Intake Folder)

Symptom:

A CSV file remains in the intake folder for more than 15 minutes and no new leads appear in the CRM.

Possible Causes:

The intake workflow is inactive or failed during execution.

The file format is invalid (non-CSV or corrupted).

Required columns are missing.

OAuth connection to Google Drive has expired.

Diagnosis Steps:

Open n8n and verify that the CSV Import workflow is active.

Check the latest execution logs for errors.

Confirm the file extension is .csv and open the file to validate structure.

Check the Error Logs sheet for recent entries related to the intake workflow.

Verify Google Drive OAuth connection status.

Fix:

If the workflow is paused, reactivate it and manually trigger execution.

If file structure is incorrect, correct the formatting and re-upload.

If OAuth is expired, reconnect Google Drive credentials in n8n.

If execution failed mid-process, resolve the specific node error and re-run the workflow.

Verification:

The file should move to the archive folder and new leads should appear in the CRM with status “New / Uncontacted.”

Prevention:

Validate CSV formatting before upload and include a weekly OAuth credential health check.

3.6.2 — Email Outreach Failure (Leads Not Receiving Emails)

Symptom:

Leads remain in “New / Uncontacted” status beyond one business day with no Date First Contacted populated.

Possible Causes:

Email Outreach workflow is disabled or failed.

No active subject line matches the lead’s vertical.

Template mismatch or missing placeholder field.

Gmail OAuth expired.

Diagnosis Steps:

Open n8n and confirm the Email Outreach workflow is active.

Review last execution timestamp and logs.

Open Email Subjects sheet and verify at least one subject is Active for the lead’s vertical.

Open Email Templates sheet and confirm required placeholders exist.

Check Gmail credential connection status.

Review Error Logs for workflow-specific errors.

Fix:

Reactivate workflow if paused.

Activate a valid subject line for the affected vertical.

Correct template placeholders or missing fields.

Reconnect Gmail OAuth if expired.

Verification:

After the next scheduled run (9:00 AM), confirm Date First Contacted is populated and Subject Used Count increments.

Prevention:

Weekly check for leads stuck in “New” status longer than 48 hours.

3.6.3 — Reply Classification Failure (Replies Not Being Processed)

Symptom:

Leads remain in “Awaiting Response” despite having replied via email.

Possible Causes:

Reply Classification workflow inactive.
Gmail read permission expired.
Email thread not marked unread.
Keyword rule mismatch.

Diagnosis Steps:

Check n8n and confirm Reply Classification workflow is active.
Review execution logs for Gmail API errors.
Open Gmail and verify replies are unread and not filtered by another label.
Check Error Logs sheet.

Fix:

Reactivate workflow if paused.
Reconnect Gmail OAuth.
Mark affected emails unread and manually trigger workflow.
Update keyword rules if legitimate replies are not being matched.

Verification:

Lead Status updates appropriately and Date Lead Responded is populated.

Prevention:

Daily review of unread Gmail count versus CRM status.

3.6.4 — Follow-Up Not Sending (Eligible Leads Skipped)

Symptom:

Leads older than 48 hours in “Awaiting Response” show Follow-Up Count = 0.

Possible Causes:

Follow-Up workflow disabled.
Date First Contacted missing.
Follow-Up Count already at maximum (2).
Next Follow-Up Date incorrectly set.

Diagnosis Steps:

Verify Follow-Up workflow is active.
Inspect affected lead record fields.
Confirm 48-hour threshold calculation.
Review execution logs for template errors.

Fix:

Correct missing Date First Contacted values.
Adjust incorrect Next Follow-Up Date.
Resolve template issues and re-run workflow.

Verification:

Follow-Up Count increments and email is sent.

Prevention:

Weekly audit of leads in “Awaiting Response” older than 7 days.

3.6.5 — Notion Transfer Failure (Qualified Leads Not Escalated)

Symptom:

Lead marked “Qualified – Pending Vetting” but no notification or transfer occurs.

Possible Causes:

Transfer workflow inactive.
Notion API token expired.
Gmail notification failure.

Diagnosis Steps:

Check transfer workflow execution history.
Verify Notion integration token validity.
Review Error Logs for API errors.

Fix:

Reconnect Notion token if expired.
Resolve workflow node errors.
Re-run workflow manually for affected lead.

Verification:

Qualified lead appears correctly in CRM and Sales Alert email is received.

Prevention:

Monitor execution logs for API authentication failures.

3.6.6 — Duplicate Leads Created

Symptom:

Multiple CRM entries exist for the same email or phone number.

Possible Causes:

CSV deduplication rule failed.
Zap intake OR-condition not functioning.
Manual duplicate upload.

Diagnosis Steps:

Compare duplicate records’ Email and Phone fields.
Check intake workflow execution logs.
Review Zap deduplication filter configuration.

Fix:

Merge duplicate records in CRM.
Correct deduplication condition in workflow.

Re-test with controlled upload.

Verification:

Uploading test file does not create duplicates.

Prevention:

Maintain OR-based dedupe logic (Email OR Phone match).

3.6.7 — Template or Placeholder Mismatch

Symptom:

Email send fails with placeholder or rendering error.

Possible Causes:

Template references non-existent field.

CRM field renamed.

Missing required data.

Diagnosis Steps:

Review error message in Error Logs sheet.

Open template and verify placeholders.

Check CRM field names and values.

Fix:

Correct placeholder syntax.

Restore missing field name.

Add missing required data to affected leads.

Verification:

Manual test send succeeds without errors.

Prevention:

Avoid renaming CRM fields without updating workflows.

3.6.8 — Gmail Rate Limiting or Deliverability Issue

Symptom:

Emails not sending or bounce notifications received.

Possible Causes:

Daily send threshold exceeded.

Sender reputation degradation.

Temporary Gmail API rate limit.

Diagnosis Steps:

Review Gmail sent folder.

Check for bounce messages.

Verify daily send count (max 50).

Fix:

Pause outreach temporarily.
Reduce daily limit if necessary.
Reconnect Gmail OAuth if API blocked.

Verification:

Test email sends successfully.

Prevention:

Maintain 50-email cap and 48-hour follow-up spacing.

3.6.9 — Full System Restart Procedure

Use Only If Multiple Workflows Are Failing

Procedure:

1. Pause all active workflows in n8n.
2. Verify credential connections (Gmail, Notion, Google Sheets, Drive).
3. Reconnect expired OAuth credentials.
4. Reactivate workflows in sequence:
 - a. Intake
 - b. Outreach
 - c. Reply Classification
 - d. Follow-Up
 - e. Transfer
5. Run manual test for each workflow.

Verification:

Confirm:

New lead processed

- Outreach sent
- Reply classified
- Follow-up scheduled

Prevention:

Quarterly credential audit and execution log review.

3.6 — Environment Setup & Replication

This section documents how to replicate SOPHIA in a staging or development environment. It enables a new developer to safely test changes, extend workflows, and validate integrations without impacting production.

At the time of this document, SOPHIA operates primarily in a single production environment. A dedicated staging environment is recommended and the replication procedure below should be followed before making structural changes.

3.7.1 Recommended Environment Architecture

Two environments should exist:

- Production — Live system connected to real CRM, Gmail, and live leads
- Staging — Isolated copy using test accounts and test data

Staging must not share:

- Gmail account
 - Notion database
 - Google Sheets template file
 - Google Drive intake folder
-

3.7.2 Cloning n8n Workflows

To replicate workflows:

1. Log into the production n8n instance.
2. Export each of the 11 workflows as JSON.
3. Log into the staging n8n instance.
4. Import workflow JSON files.
5. Rename workflows with suffix “– STAGING”.
6. Ensure all workflows remain inactive until credential updates are completed.

After import, manually review:

- Cron expressions
 - Credential references
 - Google Sheet IDs
 - Notion Database IDs
 - Webhook URLs
-

3.7.3 Setting Up Staging Integrations

Each integration must use separate test credentials.

Gmail (Staging)

- Create a dedicated test Gmail account.
- Connect OAuth2 inside staging n8n.
- Ensure daily send limit is reduced (e.g., 5 emails/day).
- Use only internal test email addresses.

Notion (Staging)

- Duplicate the production CRM database.

- Rename to “SOPHIA – Staging CRM”.
- Generate a new Notion integration token.
- Connect staging n8n to the staging database only.

Google Sheets (Staging)

- Duplicate production template file.
- Rename to “SOPHIA – Templates (Staging)”.
- Update Sheet IDs inside staging workflows.

Google Drive (Staging)

- Create new intake and archive folders.
- Update folder IDs in staging workflows.

Zapier (Staging)

- Duplicate production Zaps.
- Replace database and folder references with staging equivalents.
- Disable production Zaps while testing to prevent duplication.

3.7.4 Webflow Webhook (Staging)

If Webflow staging environment exists:

1. Duplicate the form.
2. Generate a new Zapier webhook.
3. Replace webhook URL in staging form.
4. Submit test entry and verify CRM creation.

If no Webflow staging exists, simulate webhook payload using:

- Postman
- n8n Webhook test node

3.7.5 Test Data Setup

Create structured test records representing:

- Valid new lead
- Missing placeholder field
- Lead with vertical mismatch
- Lead eligible for follow-up
- Positive reply
- Unsubscribe reply
- Bounce reply

Ensure test email inbox is monitored to validate classification logic.

3.7.6 Environment Variables & Configuration Differences

The following must differ between Production and Staging:

No workflow logic should differ between environments. Only credentials and identifiers change.

3.7.7 Validation Checklist Before Activating Staging

Before activating workflows in staging:

- All credentials connected successfully
- All folder IDs updated
- All database IDs updated
- Test email successfully sent
- Test reply classified correctly
- Follow-up triggered in controlled test

Only after validation should cron triggers be enabled.

3.7.8 Recommendation: Formal Staging Policy

- It is strongly recommended that:
- All template edits be validated in staging first.
- All schema changes be tested in staging.
- Reply classification rule updates be regression-tested.
- Production workflows never be edited directly without backup export.

Appendices

Appendix A — Document Completion Tracker

Use this tracker to monitor progress on the handover document across all three layers.

Section	Drafted	Reviewed	Approved	Owner
L1: 1.1 What SOPHIA Is	☐	☐	☐	[Name]
L1: 1.2 Business Outcomes	☐	☐	☐	[Name]
L1: 1.3 Lead Journey	☐	☐	☐	[Name]
L1: 1.4 Responsibilities	☐	☐	☐	[Name]
L1: 1.5 Warning Signs	☐	☐	☐	[Name]
L1: 1.6 Glossary	☐	☐	☐	[Name]
L2: 2.1 Daily Checklist	☐	☐	☐	[Name]
L2: 2.2 All SOPs	☐	☐	☐	[Name]
L2: 2.3 Gmail Labels	☐	☐	☐	[Name]
L2: 2.4 Lead Statuses	☐	☐	☐	[Name]
L2: 2.5 Escalation Matrix	☐	☐	☐	[Name]
L3: 3.1 Architecture	☐	☐	☐	[Name]
L3: 3.2 Workflow Registry (all 11)	☐	☐	☐	[Name]
L3: 3.3 Classification Logic	☐	☐	☐	[Name]
L3: 3.4 Database Schema	☐	☐	☐	[Name]
L3: 3.5 Integrations	☐	☐	☐	[Name]
L3: 3.6 Recovery Playbook	☐	☐	☐	[Name]
L3: 3.7 Environment Setup	☐	☐	☐	[Name]

Appendix B — Recommended Diagrams

The following visual diagrams should accompany the handover document. They can be created using Miro, Lucidchart, draw.io, or Figma.

Diagram	Purpose	Included In Layer(s)
End-to-End Lead Lifecycle	Visual flow from intake to close	Layer 1 + Layer 2
System Architecture Map	All tools and connections between them	Layer 1 (simplified) + Layer 3 (detailed)
Data Flow Diagram	How data moves between Sheets, Notion, Gmail, Drive	Layer 3
Workflow Dependency Map	Which workflows depend on or trigger others	Layer 3
Gmail Label Decision Tree	When each label is applied	Layer 2 + Layer 3
Lead Status State Machine	All possible status transitions	Layer 2 + Layer 3

Appendix C — Handover Quality Checklist

Before delivering the final handover document, verify the following:

✓	Quality Check
☐	Layer 1 is readable by a non-technical person with no prior SOPHIA knowledge
☐	Layer 1 contains NO code, workflow IDs, node names, or tool-specific jargon
☐	Layer 2 contains step-by-step numbered procedures for every recurring task
☐	Layer 2 defines what “normal” looks like so the operator can spot abnormalities
☐	Layer 3 contains exact workflow IDs, URLs, schedule times — no placeholders or example values
☐	Layer 3 documents the WHY behind design decisions, not just the WHAT
☐	All cross-references between layers are accurate (section numbers match)
☐	Google Sheets vs. Notion authority is explicitly stated for every data domain
☐	All writing guidance boxes (blue) have been removed from the final version
☐	All example boxes (yellow) have been replaced with real content and removed
☐	The document has been reviewed by someone who did NOT build the system
☐	Credential locations are referenced but actual passwords/keys are NOT in the document